

OKAYAMA, Japan

WMO: 477680

Lat: 34.6858N

Lon: 133.9253E

Elev: 7

StdP: 101.24

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	-0.3	-8.5	1.8	4.2	-7.1	2.1	4.6	12.1	4.9	10.7	5.4	2.9	270	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	34.6	25.4	33.6	25.1	32.6	24.8	26.3	32.1	25.9	31.3	25.5	30.6	4.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.0	28.5	24.5	19.5	28.5	24.1	19.0	28.4	82.0	32.3	80.1	31.3	78.4	30.7	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.7	8.3	7.0	-3.1	36.2	1.2	1.0	-4.0	36.9	-4.8	37.5	-5.4	38.0	-6.3	38.7		
			-4.4	27.1	1.2	0.6	-5.3	27.6	-6.0	28.0	-6.8	28.3	-7.7	28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	5.1	5.8	9.1	14.6	19.8	23.5	27.8	28.9	24.8	18.8	12.7	7.4
	DBStd	8.66	2.13	2.60	2.98	3.20	2.36	2.33	2.28	1.81	2.82	2.93	3.06	2.58
	HDD10.0	431	153	121	55	4	0	0	0	0	0	9	88	
	HDD18.3	1720	412	351	288	117	13	0	0	0	29	171	340	
	CDD10.0	2833	1	3	26	142	305	406	552	587	443	273	89	7
	CDD18.3	1079	0	0	0	4	59	156	293	329	193	44	1	0
	CDH23.3	10413	0	0	0	32	339	1035	3205	4016	1616	169	0	0
	CDH26.7	3989	0	0	0	2	48	264	1337	1836	487	16	0	0
Wind	WSAvg	2.9	3.3	3.1	3.2	3.1	2.9	2.8	2.9	3.0	2.8	2.5	2.6	3.1
Precipitation	PrecAvg	1177	39	50	83	88	120	169	167	104	152	95	64	46
	PrecMax	1646	112	104	195	188	305	385	372	306	448	269	151	104
	PrecMin	791	6	16	34	31	24	24	41	18	50	25	12	5
	PrecStd	237	25	23	36	38	61	82	90	77	98	63	39	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.5	16.5	20.7	26.2	29.9	33.1	35.6	36.1	33.7	28.5	22.1	16.3
		MCWB	8.7	11.8	12.6	15.8	18.0	23.7	25.8	25.8	24.8	22.0	16.4	11.8
	2%	DB	11.7	14.0	18.4	24.0	28.0	30.7	34.4	35.0	31.9	26.5	20.7	14.9
		MCWB	7.5	9.4	11.7	14.9	17.6	22.3	25.4	25.5	24.2	20.3	15.6	11.0
	5%	DB	10.6	12.3	16.5	22.2	26.6	29.4	33.4	34.0	30.6	25.2	19.4	13.5
		MCWB	6.5	8.0	10.7	14.3	17.1	21.3	25.1	25.3	23.5	19.2	14.8	9.4
	10%	DB	9.4	10.7	14.8	20.6	25.3	28.1	32.3	33.0	29.5	23.9	18.0	12.1
		MCWB	5.6	6.7	9.8	13.5	16.9	20.8	24.9	25.0	22.9	18.1	13.7	8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	13.3	14.9	19.1	21.8	25.0	27.0	27.1	26.0	23.8	18.7	14.2
		MCDB	12.2	15.9	17.6	22.3	24.9	30.1	33.7	33.4	31.0	26.7	20.0	15.4
	2%	WB	8.3	10.5	13.4	17.5	20.4	24.1	26.2	26.5	25.2	22.0	17.0	11.6
		MCDB	10.5	12.9	16.2	20.4	23.7	28.5	32.1	32.4	29.6	25.4	19.3	13.7
	5%	WB	7.2	8.8	12.1	16.2	19.4	23.4	25.8	26.1	24.7	20.6	15.8	10.0
		MCDB	9.8	11.3	15.3	20.0	23.2	27.3	31.2	31.6	28.9	23.8	18.4	12.5
	10%	WB	6.0	7.2	10.5	15.1	18.5	22.6	25.4	25.7	24.2	19.2	14.4	8.8
		MCDB	8.9	10.1	14.2	19.0	22.7	26.1	30.5	30.9	28.2	22.7	17.3	11.6
Mean Daily Temperature Range	5% DB	MDBR	7.5	7.9	8.9	9.4	9.3	7.5	7.1	7.3	7.1	7.9	7.9	7.6
		MCDBR	8.9	9.7	11.1	11.9	11.8	9.5	8.7	8.6	7.8	8.5	8.6	8.9
		MCWBR	5.8	6.3	6.6	5.4	4.7	3.3	2.6	2.6	2.9	4.0	5.0	5.7
	5% WB	MCDBR	8.1	8.3	9.1	8.4	8.1	6.9	7.9	7.7	6.7	6.6	7.1	7.8
		MCWBR	6.0	6.3	6.8	5.1	4.6	2.9	2.6	2.5	2.8	4.2	5.0	5.8
Clear-Sky Solar Irradiance	taub	0.416	0.461	0.528	0.557	0.559	0.581	0.581	0.552	0.503	0.457	0.432	0.409	
	taud	2.086	1.987	1.844	1.831	1.887	1.908	1.961	2.028	2.126	2.163	2.133	2.124	
	Ebn at Noon	754	762	745	748	753	734	731	745	761	761	733	734	
	Edh at Noon	127	156	196	208	199	195	184	169	146	130	121	116	
All-Sky Solar Radiation	RadAvg	2.27	2.90	3.91	4.84	5.29	4.80	5.04	5.20	4.04	3.33	2.46	2.02	
	RadStd	0.16	0.24	0.30	0.42	0.49	0.42	0.67	0.53	0.40	0.32	0.20	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page